

1-YEAR AI LEARNING PLAN

A structured weekly breakdown

This document outlines a comprehensive 1-year AI learning plan, designed for individuals with a foundational understanding of programming. This plan assumes a time commitment of approximately **2 hours per day, 5 days per week**, totaling **10 hours weekly**. The recommended study days are **Monday to Friday**, with **Saturday and Sunday** reserved for breaks, revision, and experimentation.

Each phase will be broken down into weekly topics and mini-projects to help re-inforce the learning.

Phase 1: Python Fundamentals & Essential Libraries (Weeks 1-8)

- **Duration:** 8 Weeks
- **Focus:** Building a strong foundation in Python and essential libraries for AI.

Week 1:

- Python Basics
- Data Types
- Variables

Week 2:

- Control Flow
- Loops
- Functions

Week 3:

- Data Structures
- Lists, Tuples
- Dictionaries, Sets

Week 4:

- Object-Oriented Programming (OOP) - Basics
- Classes
- Objects

Week 5:

- NumPy
- Arrays
- Operations

Week 6:

- Pandas
- DataFrames
- Data Manipulation

Week 7:

- Matplotlib
- Data Visualization
- Plots and Charts

Week 8:

- Seaborn
- Advanced Visualizations

Mini-Projects/Practice:

- **Week 1:** Create a simple calculator.
- **Week 2:** Build a number guessing game.
- **Week 3:** Implement a basic to-do list.
- **Week 4:** Design a simple class representing a or .

Mini-Projects/Practice:

- **Week 5:** Perform array operations using NumPy.
- **Week 6:** Analyze a dataset using Pandas.
- **Week 7:** Create different types of graphs using Matplotlib.
- **Week 8:** Improve visualizations using Seaborn. (e.g. create a heatmap)

Phase 2: Machine Learning Fundamentals (Weeks 9-24)

- **Duration:** 16 Weeks
- **Focus:** Diving into core Machine Learning concepts and algorithms.

Week 9:

- Introduction to Machine Learning
- Types of ML
- Supervised Learning

Week 10:

- Unsupervised Learning
- Reinforcement Learning
- Bias-Variance Tradeoff

Week 11:

- Linear Regression
- Model Evaluation
- Cost Functions

Week 12:

- Multiple Linear Regression
- Polynomial Regression
- Feature Scaling

Mini-Projects/Practice:

- **Week 9:** Implement a simple supervised learning algorithm.
- **Week 10:** Explore a dataset and identify potential unsupervised learning tasks.
- **Week 11:** Build a simple linear regression model to predict housing prices.
- **Week 12:** Expand the regression model to handle multiple features.

Week 13:

- Logistic Regression
- Classification Metrics
- Threshold Tuning

Week 14:

- K-Nearest Neighbors (KNN)
- Distance Metrics
- Hyperparameter Tuning

Week 15:

- Decision Trees
- Information Gain
- Tree Pruning

Week 16:

- Random Forests
- Ensemble Methods
- Feature Importance

Week 17:

- Support Vector Machines (SVM)
- Kernels
- Regularization

Week 18:

- Clustering Algorithms
- K-Means
- Hierarchical Clustering

Week 19:

- Dimensionality Reduction
- Principal Component Analysis (PCA)
- Feature Selection

Week 20:

- Model Selection
- Cross-Validation
- Grid Search

Mini-Projects/Practice:

- **Week 13:** Create a logistic regression model for classifying emails as spam or not spam.
- **Week 14:** Build a KNN classifier for image recognition.
- **Week 15:** Implement a decision tree for predicting customer churn.
- **Week 16:** Construct a random forest model for a classification task.

Mini-Projects/Practice:

- **Week 17:** Build an SVM classifier for a dataset of your choice.
- **Week 18:** Apply K-Means clustering to segment customers based on their purchasing behavior.
- **Week 19:** Use PCA to reduce the dimensionality of a high-dimensional dataset.
- **Week 20:** Compare the performance of different models using cross-validation.

Week 21:

- Introduction to Neural Networks
- Perceptron
- Activation Functions

Week 22:

- Multi-Layer Perceptron (MLP)
- Backpropagation
- Optimization Algorithms

Week 23:

- Regularization Techniques
- Dropout
- Early Stopping

Week 24:

- Introduction to Deep Learning Frameworks
- TensorFlow
- Keras

Mini-Projects/Practice:

- **Week 21:** Implement a single-layer perceptron for binary classification.
- **Week 22:** Build a multi-layer perceptron to classify handwritten digits (MNIST).
- **Week 23:** Experiment with regularization techniques to prevent overfitting in a neural network.
- **Week 24:** Get familiar with TensorFlow and Keras by implementing a simple model.

Phase 3: Advanced Machine Learning & Deep Learning (Weeks 25-40)

- **Duration:** 16 Weeks
- **Focus:** Exploring advanced ML techniques and deep learning architectures.

Week 25:

- Convolutional Neural Networks (CNNs)
- Convolutional Layers
- Pooling Layers

Week 26:

- CNN Architectures
- Image Classification
- Object Detection

Week 27:

- Recurrent Neural Networks (RNNs)
- LSTM
- GRU

Week 28:

- Sequence Modeling
- Text Generation
- Machine Translation

Mini-Projects/Practice:

- **Week 25:** Build a CNN to classify images from the CIFAR-10 dataset.
- **Week 26:** Implement object detection using a pre-trained CNN model.
- **Week 27:** Build an RNN to predict the next character in a text sequence.
- **Week 28:** Develop a simple machine translation model using RNNs.

Week 29:

- Autoencoders
- Dimensionality Reduction
- Anomaly Detection

Week 30:

- Generative Adversarial Networks (GANs)
- Generators
- Discriminators

Week 31:

- Reinforcement Learning Algorithms
- Q-Learning
- SARSA

Week 32:

- Deep Reinforcement Learning
- DQN
- Policy Gradients

Mini-Projects/Practice:

- **Week 29:** Build an autoencoder to compress images.
- **Week 30:** Train a GAN to generate realistic images.
- **Week 31:** Implement Q-learning to train an agent to play a simple game.
- **Week 32:** Use DQN to train an agent to play Atari games.

Week 33:

- Transfer Learning
- Pre-trained Models
- Fine-tuning

Week 34:

- Natural Language Processing (NLP)
- Text Preprocessing
- Word Embeddings

Week 35:

- Sentiment Analysis
- Text Classification
- Named Entity Recognition

Week 36:

- Topic Modeling
- Latent Dirichlet Allocation (LDA)
- Non-negative Matrix Factorization (NMF)

Week 37:

- Time Series Analysis
- ARIMA
- LSTM for Time Series

Week 38:

- Graph Neural Networks (GNNs)
- Node Embeddings
- Graph Classification

Week 39:

- Bayesian Machine Learning
- Bayesian Linear Regression
- Gaussian Processes

Week 40:

- Causal Inference
- Counterfactuals
- Intervention Analysis

Mini-Projects/Practice:

- **Week 33:** Fine-tune a pre-trained CNN model for a specific image classification task.
- **Week 34:** Preprocess text data and create word embeddings using Word2Vec or GloVe.
- **Week 35:** Build a sentiment analysis model to classify movie reviews as positive or negative.
- **Week 36:** Apply topic modeling to discover the main topics in a collection of news articles.

Mini-Projects/Practice:

- **Week 37:** Forecast future stock prices using ARIMA or LSTM.
- **Week 38:** Build a GNN to classify nodes in a social network.
- **Week 39:** Implement Bayesian linear regression to predict house prices with uncertainty estimates.
- **Week 40:** Analyze the causal effect of a marketing campaign on sales using causal inference techniques.

Phase 4: Specialization & Projects (Weeks 41-52)

- **Duration:** 12 Weeks
- **Focus:** Deepening knowledge in a specific AI area and completing significant projects.

This phase focuses on project-based learning. Choose a specific area of AI that interests you (e.g., Computer Vision, NLP, Reinforcement Learning, etc.) and work on 2-3 substantial projects to solidify your skills. Examples include:

- **Computer Vision:** Building an image recognition system, object detection model, or image segmentation tool.
- **Natural Language Processing:** Creating a chatbot, sentiment analysis tool, or text summarization system.
- **Reinforcement Learning:** Training an agent to play a complex game or control a simulated robot.

During this phase, spend time reading research papers, contributing to open-source projects, and networking with other AI practitioners. Document your projects thoroughly and showcase them on platforms like GitHub or personal websites.

Sample Weekly Study Routine

This table provides an example of how to structure your weekly study routine. Adjust the timings and topics based on your individual needs and the current phase of the learning plan.

Day	Time	Task
Monday	6:00 PM - 8:00 PM	Study new concepts/theory
Tuesday	6:00 PM - 8:00 PM	Work on mini-project / practice problems
Wednesday	6:00 PM - 8:00 PM	Review previous week's material

Thrusday	6:00 PM - 8:00 PM	Read research papers / articles
Friday	6:00 PM - 8:00 PM	Coding challenge / problem solving
Saturday	6:00 PM - 8:00 PM	Flexible Revision / Experimentation
Sunday	6:00 PM - 8:00 PM	Break / Relax

Remember to be consistent, stay curious, and enjoy the journey of learning AI! Good luck!